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MILITARY SPECIFICATION

ADHESIVES, FIRE-RESISTANT, THERMAL INSULATION

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope. This specification covers fire-resistant adhesives for the application and repair of fibrous glass insulation board, on piping and machinery and other metal surfaces.

1.2 Classification. The adhesives shall be of the following types, as specified (see 6.1):

Type I —For use in repairing damaged surface of faced fibrous glass insulation board and for securing cotton brattice cloth to unfaced fibrous glass insulation board.

Type II —For securing fibrous glass tape to the joints of faced fibrous glass insulation board and securing lagging cloth to thermal insulation on piping and machinery.

Type III—For securing bonded fibrous glass insulation to metal surfaces.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

TT–E–485 —Enamel, Semi-Gloss, Rust Inhibiting.

PPP–C–96 —Cans, Metal, 28 Gage and Lighter.

PPP–D–760 —Drums and Pails, Metal (5 and 16.64 Gallon).

MILITARY

MIL–I–742 —Insulation Board, Thermal, Fibrous Glass.

MIL–C–788 —Cloth, Brattice, Cotton, Fire-Resistant.

MIL–P–17970 —Paint, Nonflaming (Dry) Chlorinated Alkyd Resin, Soft White, Semigloss, Formula No. 124/58.

MIL–C–20079 —Cloth, Glass; Tape, Textile, Glass; and Thread, Glass.

MIL–I–22023 —Insulation Felt, Thermal and Sound Absorbing Felt, Fibrous Glass, Flexible.

STANDARDS

FEDERAL

FED–STD–141—Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.

FED–STD–595—Colors.

FED–STD–791—Lubricants, Liquid Fuels, and Related Products; Methods of Testing.

MILITARY

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

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(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

INTERSTATE COMMERCE COMMISSION

49 CFR 71–78—Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(These regulations are now a part of the Code of Federal Regulations available from the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Applications for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N. Y.)

3. REQUIREMENTS

3.1 Qualification. The adhesive furnished under this specification shall be a product which has been tested, and passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable qualified products list.

3.2 Material. The adhesives shall be effective for the purpose intended without heating or the addition of other ingredients. They shall be free of all ingredients which may adversely affect the serviceability or have a deleterious effect on thermal insulation and cotton brattice cloth. They shall be free from grit, lumps, and skins.

3.3 Stability.

3.3.1 The adhesives shall not liver, settle hard, or otherwise deteriorate when stored for a period of 6 months in airtight containers (see 4.4.1.1).

3.3.2 The adhesives shall be stable and not be adversely affected on returning to room temperature after subjection to a temperature of 16° Fahrenheit (F.) (see 4.4.1.2).

3.4 Toxicity and irritancy. The adhesives shall be nontoxic and nonirritating to the workmen using them.

3.5 Fire resistance. There shall be no residual flame or continuous burning of any specimen after removal of the test flame when adhesives are tested as specified in 4.4.2.

3.6 Flexibility. Adhesives shall not chip, scale, or otherwise leave the surface of the tin strip when tested as specified in 4.4.3.

3.7 Type I adhesive.

3.7.1 *Consistency.* The consistency of type I adhesive at 80° F., after it has been stirred moderately by hand, shall be suitable for brush application of the surface of faced and unfaced fibrous glass insulation board conforming to Specification MIL–I–742 and to the surface of cotton brattice cloth conforming to Specification MIL–C–788 (see 4.4.4).

3.7.2 *Flash point.* The flash point of the adhesive shall be not less than 110°F. (see 4.4.5).

3.7.3 *Initial strength.* The initial strength of the adhesive shall be sufficient to secure cotton brattice cloth to fibrous glass insulation board mounted in a vertical or overhead position without the aid of shoring, pinning, or other mechanical devices (see 4.4.6).

3.7.4 *Coverage and finished weight.* The coverage obtained from the adhesive in producing a satisfactory covering shall not be less than 30 square feet per gallon when determined as specified in 4.4.7. The finished weight of the covering shall be not more than 5.35 ounces per square foot of application.

3.7.5 *Drying time and adhesive strength.* The adhesive shall dry to touch within a period of 4 hours and dry thoroughly to constant weight within a period of 48 hours, and shall have an average adhesive strength of not less than 3 pounds when tested as specified in 4.4.8.

3.7.6 *Color.* The adhesive when applied and dried shall be white in color (see 4.4.9).

3.7.7 *Washability.* The adhesive shall be washable to the extent that there shall be no definite break through or detachment of the adhesive when tested as specified in 4.4.10.

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3.7.8 *Paintability.* The adhesive shall not cause discoloration or bleeding through paint conforming to Specification MIL–P–17970, shall not cause cracking or crazing of the paint, and shall be compatible with the paint when tested as specified in 4.4.11.

3.7.9 *Fire resistance of the finished covering.* The finished covering shall be rated as noncombustible or fire-retardant when tested as specified in 4.4.12.

3.7.10 *Puncture resistance.* The puncture resistance of the finished covering shall be not less than 800 inch-ounces per inch of tear when tested as specified in 4.4.13.

3.8 Type II adhesive.

3.8.1 *Viscosity.* The viscosity of the adhesive at 77°F. shall be not less than 100 nor more than 110 Krebs units (see 4.4.14).

3.8.2 *Flash point.* The flash point of the adhesive shall be not less tran 100°F. (see 4.4.5).

3.8.3 *Adhesive strength.* The adhesive strength before and after aging shall be such that a specimen will support a 2,000-gram weight for a period of 10 minutes (see 4.4.15.1).

3.8.4 *Paintability.* The adhesive shall not cause discoloration or bleeding through paint conforming to Specification MIL–P–17970, shall not cause cracking or crazing of the paint, and shall be compatible with the paint when tested as specified in 4.4.11.

3.9 Type III adhesive.

3.9.1 *Protection against corrosion.* Adhesive shall protect bare steel from corrosion when tested as specified in 4.4.16.

3.9.2 *Viscosity.* The viscosity of the adhesive at 77°F. shall be not less than 100 nor more than 130 Krebs units, when tested as specified in 4.4.14.

3.9.3 *Adhesive strength.* The adhesive strength before and after aging shall be such that a specimen will support a 2,000-gram weight for a period of 10 minutes when tested as specified in 4.4.15.2.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Qualification tests. Qualification tests shall be conducted at a laboratory satisfactory to the Bureau of Ships and shall consist of the tests specified in 4.4.

4.3 Acceptance inspection.

4.3.1 *Lot.* For purpose of acceptance inspection, a lot shall consist of all adhesive from one production batch.

4.3.2 *Examination of filled containers.* A random sample of filled containers from each inspection lot shall be examined in accordance with Standard MIL–STD–105 at inspection level I and acceptable quality level equal to 2.5 percent defective to verify compliance with this specification. Any container in the sample having one or more defects or under required fill shall be rejected, and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of Standard MIL–STD–105, the lot represented by the sample shall be rejected.

4.3.3 *Acceptance tests.* From each inspection lot two containers shall be taken. From each of the containers, two-quart samples shall be subjected separately to the tests specified in 4.4.3., 4.4.5, 4.4.8, 4.4.15.1, and 4.4.15.2 as applicable to the types. If any one of the samples tested is found to be not in conformance with this specification, the lot which it represents shall be rejected.

4.4 Test procedures.

4.4.1 *Stability.*

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4.4.1.1 The adhesive shall be stored for a period of 6 months in an airtight container. At the end of this period, the adhesive shall be examined to determine conformance with 3.3.1.

4.4.1.2 The adhesive shall be subjected to a temperature of 16°F. for 16 hours before testing.

4.4.2 *Fire resistance.* Strips of fibrous glass tape, 6 inches long and 2 inches wide, conforming to type II, class 1 of Specification MIL–C–20079, shall be coated with the adhesive. The coated strips shall be air dried for 24 hours and then placed in an electrically heated oven for 20 hours at 212° to 221°F. The specimens shall then be removed from the oven and individually suspended vertically from a clamp covering the upper ½ inch of the strip. A Meker burner, with a 30 millimeter diameter grid, shall then be placed below the specimen in such a position that ½ inch of the specimen is in the 1½-inch test flame. The burner shall be removed at the end of 5 seconds and the length of time the coating burned noted.

4.4.3 *Flexibility.*

4.4.3.1 *Apparatus.* The apparatus shall consist of the following:

A panel 8 by 10 inches, 0.011 inch thick, made from bright tin-plate.

A film applicator with a clearance of 0.02 inch, and a gap of sufficient size to make a film 3 to 6 inches wide.

A steel mandrel ¼ inch in diameter and 6 inches long supported at each end.

4.4.3.2 *Procedure.* The 8 by 10 inch tin panel shall be placed on a flat surface. Using the film applicator, a film of the adhesive 10 inches long and 3 to 6 inches wide shall be applied on the panel. The coated panel shall be airdried for 24 hours. Five 1½ by 8-inch strips shall be cut from the coated panel. The strips shall be placed in an electric air-circulating oven and baked at a temperature of 212 to 221°F. for 75 minutes. The strips shall be removed from the oven and allowed to cool to room temperature. Then the strips shall be bent rapidly over the surface of the mandrel. The strips shall be examined for cracks at the bend. Slight surface cracks shall be disregarded, but deep cracks which expose bare metal shall be considered as failure of the adhesive to pass the test.

4.4.4 *Consistency.* Type I adhesive, after it has been stirred moderately by hand, shall be applied with a brush to faced and unfaced fibrous glass insulation board conforming to Specification MIL–I–742 and to the surface of cotton brattice cloth conforming to Specification MIL–C–788.

4.4.5 *Flash point.* The flash point of type I and type II adhesives shall be determined in accordance with method 1103 of Standard FED–STD–791.

4.4.6 *Initial strength.* A prime or underlying coat of type I adhesive shall be applied to 2 feet by 3 feet sections of faced and unfaced fibrous glass insulation board conforming to Specification MIL–I–742, mounted in a vertical position. The adhesive shall be allowed to set for 5 minutes. Two foot by three foot pieces of cotton brattice cloth conforming to Specification MIL–C–788 shall be superimposed upon the primed surfaces and hand troweled smooth, eliminating air pockets. An overlying or finish coat of the adhesive shall then be brushed over the cotton brattice cloth.

4.4.7 *Coverage and finished weight.* A prime or underlying coat of type I adhesive shall be applied to the surface of 2-foot by 3-foot sections of faced and unfaced fibrous glass insulation board conforming to Specification MIL–I–742. Two-foot by three-foot pieces of cotton brattice cloth conforming to Specification MIL–C–788 shall then be immediately superimposed upon the primed surfaces and hand troweled smooth, eliminating air pockets. An overlying or finish coat of adhesive shall then be brushed over the cotton cloth. The specimens shall be weighed before and after each application of the adhesive. Coverage shall be based on the difference in weight before and after application of the adhesive and the specific gravity of the coating, and shall be reported as the average square feet per gallon for both faced and unfaced board.

4.4.8 *Drying time and adhesive strength.* Specimens of 12- by 12- by 1-inch sections of faced fibrous glass insulation board, conforming to Specification MIL–I–742 shall be coated type I with adhesive to an approximate thickness of 1/32 inch. After the adhesive has dried for 5 minutes, 6- by 12-inch strips of cotton brattice cloth, in accordance with Specification MIL–C–

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788, shall be applied in such a manner that a strip of cloth covers one-half of the coated section. The cloth shall be pressed and troweled until the adhesive extrudes through the interstices and then shall be given a finish coating. The specimens shall then be allowed to dry for 48 hours at approximately 80°F. room temperature. At the end of this time, three 2- by 6-inch strips of the covering shall be cut and removed from the surface of the fibrous glass insulation board. The plies of the strips thus obtained shall be stripped back 1 inch by hand, and then pulled to rupture in a tensile testing machine, the jaws of which separate at the rate of 12 inches per minute. The average results obtained from three strips of specimens shall be reported as the adhesive strength.

4.4.9 *Color.* The specimens prepared for the test specified in 4.4.8 shall be examined to determine the color of the adhesive when dried.

4.4.10 *Washability.*

4.4.10.1 *Apparatus.* The apparatus shall consist of a heavy duty winshield wiper or any other suitable form of propulsion mounted on or under a metal plate in such a manner that it will pull a brush back and forth across a test panel fastened to the metal plate. The brush shall be mounted on the apparatus so that its long axis shall be parallel to the direction of movement of the brush. The block of the brush shall be close-grained hardwood or aluminum, 3¾ inches long, 1½ inches wide, and 1 inch thick, free from all defects. The brush stock shall be stiff, black butt-cut chinese hog bristle. There shall be 60 holes in the block 5/32 inch in diameter, solidly filled with bristle. The bristles shall extend ¾ inch beyond the block. The bristle ends shall be dressed down with sandpaper, or burned on a hot plate, if necessary, so that the abrading surface of the brush is as nearly plane as possible. Symmetrically placed weights shall be fastened to the brush to bring its total weight to one pound.

4.4.10.2 *Preparation of test panel.* A film of type I adhesive shall be drawn down on a 10 by 18-inch plate glass panel with a film applicator of 0.02 inch clearance. The draw down shall be 3½ inches or more in width and shall be made parallel to the long dimension of the panel with an even, fairly rapid motion. The coated panel shall be allowed to airdry for 6 days at room temperature.

4.4.10.3 *Procedure.* The brush bristles shall be immerced in water at 25° to 30°C. for 30 minutes to a depth of ½ inch. Then cake grit soap shall be rubbed into the brush for about 30 seconds, working the soap well up into the bristles. The brush should be turned on its back during this operation so that no more than a minimum amount of water is lost. The test panel shall be mounted firmly on the washability apparatus, the brush shall be placed on the adhesive film and started oscillating immediately. The speed of the brush shall be adjusted to approximately 40 oscillations a minute. During the test, water at a rate of about 12 drops per minute or just sufficient to keep the panel wet shall be allowed to drop or run into the path of the brush. The panel shall be removed from the apparatus at the end of 1,000 brush oscillations and washed immediately in water at moderate temperature. The film shall be inspected within the middle 4 inches of brush travel by both reflected and transmitted light for definite breaks, wear, or detachment of the film down to the glass. A few small "pin point" size areas may be disregarded.

4.4.11 *Paintability.*

4.4.11.1 *Type I adhesive.* The adhesive shall be applied to the facing of a 12 inch by 12 inch specimen of fibrous glass insulation board conforming to Specification MIL–I–742. A 12 inch by 12 inch piece of cotton brattice cloth, conforming to MIL–C–788, shall be superimposed upon the primed facing and hand troweled smooth, eliminating and removing air pockets, until the adhesive extrudes through the interstices of the cloth. An overlying or finish coat of the adhesive shall then be brushed over the cloth. The specimen shall be allowed to airdry for 48 hours. A coat of paint conforming to Specification MIL–P–17970 shall be applied to the face of the specimen and allowed to airdry for 24 hours. Then a second coat of paint shall be applied. The specimen shall be inspected after 24 and 48 hours for signs of discoloration, bleeding, cracking or crazing of the paint.

4.4.11.2 *Type II adhesive.* The adhesive

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shall be applied to the glass cloth facing of a 12 inch by 12 inch specimen of fibrous glass insulation board conforming to Specification MIL–I–742 and allowed to set for 10 minutes. A strip of fibrous glass tape conforming to type II, class 1 of Specification MIL–C–20079, shall then be applied to the adhesive-coated specimen. Then a coat of adhesive shall be applied over the tape. The specimen shall be allowed to airdry for 24 hours. A coat of paint conforming to Specification MIL–P–17970 shall be applied to the face of the specimen and allowed to airdry for 24 hours after which a second coat of paint shall be applied. The specimen shall be inspected after 24 and 48 hours for signs of discoloration, bleeding, cracking or crazing of the paint.

4.4.12 *Fire resistance of the finished covering.*

4.4.12.1 Type I adhesive shall be applied to 24 by 30 by 1-inch specimens of type I and type II fibrous glass insulation boards, conforming to Specification MIL–I–742. Then 24 by 30-inch pieces of cotton brattice cloth conforming to MIL–C–788, shall be superimposed upon the primed surfaces of the specimens and hand troweled smooth, eliminating and removing air pockets, until the adhesive extrudes through the interstices of the cloth. An overlying or finish coat of adhesive shall then be brushed over the cloth. Then the specimens shall be applied directly to noncombustible backings. Before the test each specimen shall be dried to constant weight at a temperature not injurious to the material being tested. Each specimen shall be tested separately.

4.4.12.2 The test shall be conducted in a room which is free from appreciable air currents and which has a temperature between 60° and 85° F. The specimen shall be placed in a horizontal position with the surface to be exposed to the fire facing downward, and shall be supported on the flat surface of a by 2 by ⅛-inch steel angle frame. (Standard laboratory equipment has a 30 by 30-inch clear opening.) The flaflme from a ¾ to ⅞-inch gas-air burner shall be directed against the center of the lower surface of the specimen. The top of the burner shall be directed against the center of the lower surface of the specimen. The top of the burner tube shall be 28¾ inches below the specimen. Temperature indications shall be obtained with a chromel-alumel thermo element made of 0.12849-inch wire placed in a 3-inch horizontal coil 1 inch below the center of the specimen. The wires shall be bare for a distance of 2 inches from the junction. Temperature readings shall be taken at intervals not exceeding 2 minutes. The test duration shall be 40 minutes and the flame shall be regulated to give temperature indications according to the time temperature curve shown on figure 1. The flame shall touch the specimen during the entire test period. Exceptions can be made for the first 5 minutes, if required, for proper temperature regulation. At no time during the test shall the flame cover a greater area on the specimen than a 12-inch diameter circle.

4.4.12.3 The area under the time temperature curve obtained from the thermo element readings shall be within 5 percent of that of the reference curve being followed.

4.4.12.4 For noncombustible and fire retardant materials, the specimen as a whole and the different units or portions thereof shall remain in place until the end of the flame exposure period specified in 4.4.12.2. Exception shall be made for burned, charred, or disintegrated material falling in pieces having an area smaller than 50 square inches. No glow during or after the test shall progress to the edge of the 36 by 36-inch specimen at any point.

4.4.12.5 *Noncombustile.* When adhesive is subjected to the test specified in 4.4.12.2, no flame shall issue from the specimen during or after flame application.

4.4.12.6 *Fire retardant.* When adhesive is subjected to the test specified in 4.4.12.2, no sustained flaming shall issue from the specimen. Any flame which occurs shall be limited to intermittent short flames from the area directly exposed to the test flame. No flame from the specimen shall reach the angle frame at any point. No flaming shall occur more than 2 minutes after the test flame is discontinued.

4.4.13 *Puncture resistance.* The puncture resistance of the finished covering composed of dry coatings of the adhesive and the cotton brattice cloth laminate on the glass cloth facing of a 24 by 18 by 1-inch specimen of fibrous glass in-

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sulation board 24 by 18 inches, conforming to type I of Specification MIL–I–742, shall be determined by the method specified in 4.4.13.2.

4.4.13.1 *Apparatus.* The apparatus, shown on figure 2, for measuring the puncture resistance of the finished covering shall consist of the following:

(a) *Pendulum.* A pendulum incorporating an arm in the form of an arc.
(b) *Puncture point.* A puncture point affixed to the end of the arm and having the shape of a right angle triangle pyramid 1 inch in height.
(c) *Pointer collar.* A loose collar around the base of the puncture point to eliminate friction resulting from the punctured material rubbing against the puncture arm.
(d) *Trigger.* A trigger, or latch, to hold the pendulum in the horizontal position.
(e) *Two clamping plates.* Two clamping plates for holding the specimen to be tested.
(f) *Pointer and scale.* A pointer and scale to indicate the arc through which the pendulum travels after penetrating through the specimen.
(g) *Weights.* Weights, which, when attached, give a maximum scale value of at least 350 inch-pounds of energy when the pendulum is in horizontal position, and a scale graduation corresponding to the weights used.
(h) *Loose plate.* A loose plate which is inserted between the clamping plates when testing material for the purpose of defining the test area. The plate shall be removed when checking the zero reading of the apparatus.
(i) *Adjustable screw.* An adjustable screw bears against the pointer assembly during the measuring half of the pendulum swing, and permits adjustment of the zero position of the pointer.

4.4.13.2 *Procedure.* The test specimen shall be placed, with the finished covering surface down, between the clamping plates. Before each test is made, the loose sleeve shall be placed against the base of the puncture point and the pointer set about 1 inch above the expected reading. The pendulum shall be raised to the horizontal position and released by pushing the latch handle to the left. The reading shall be noted on the proper scale after the pendulum has completed its swing. The scales are graduated in inch-ounces per inch of tear.

4.4.14 *Viscosity.* The viscosity of type II and type III adhesives shall be determined in accordance with method 4281 of Standard FED–STD–141.

4.4.15 *Adhesive strength.*

4.4.15.1 *Type II.*

4.4.15.1.1 *Preparation of samples.* The adhesive shall be applied on the facing of 6 specimens of type I fibrous glass insulation board, 10 inches long by 3 inches wide by 1 inch thick, conforming to Specification MIL–I–742 and allowed to set for 10 minutes. Strips of fibrous glass tape plain weave 12 inches long by 2 inches wide conforming to type II, class 1 of Specification MIL–C–20079 shall then be applied to the adhesive-coated specimens. A coat of the adhesive shall be applied over the tape. The six specimens shall be allowed to dry for 24 hours. Three of the specimens shall then be aged at a temperature of 194° F. for 24 hours.

4.4.15.1.2 *Procedure.* The tape shall be stripped one inch from the end of each specimen. Each of the specimens shall be supported in a horizontal position with the tape side down. A 2000-gram weight shall be attached to the free end of the tape and suspended for a period of 10 minutes.

4.4.15.2 *Type III.*

4.4.15.2.1 *Preparation of samples.* Six bare steel plates, 10 inches long by 6 inches wide by ⅛ inch thick, shall be coated with adhesive and allowed to set for 10 minutes. Six specimens of bonded fibrous glass insulation, 12 inches long by 3 inches wide by 1 inch thick, conforming to Specification MIL–I–22023 shall then be applied to the coated steel plates and allowed to dry for 24 hours.

4.4.15.2.2 *Procedure before aging.* Three of the specimens shall be supported at each end in

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a horizontal position with bonded fibrous glass side down, a 2,000-gram weight attached to the free end of the insulation, and suspended for a period of 10 minutes.

4.4.15.2.3 *Procedure after aging.* The remaining three specimens shall be aged at a temperature of 194°F. for 24 hours, prior to testing as specified in 4.4.15.2.2.

4.4.16 *Protection against corrosion.* Ground steel strips 1/8 by 1 by 6 inches shall be dripped in type III adhesive and allowed to dry for 24 hours. At the end of this time the strips shall be subjected to a 4-percent salt solution spray for 72 hours at 100° F. The strips shall be removed from the spray and examined for penetration of the salt spray and corrosion of the metal surfaces.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 *Level A.* The adhesive shall be furnished in 1-gallon multifriction top cans conforming to type V, class 2 of Specification PPP–C–96 or 5-gallon lug cover pails conforming to type II, class 1 or 2, of specification PPP–D–760.

5.1.1.1 *One-gallon cans.* Exterior plan B coating of cans will be required, and after filling, the plugs shall be spot-soldered to the friction ring at three points equidistant from each other around the periphery of the plug.

5.1.1.2 *Five-gallon pails.* The exterior coating of pails shall conform to Specification TT–E–485. Color shall be olive drab conforming to color number 2430 of Standard FED–STD–595.

5.1.2 *Level C.* The adhesive shall be furnished in commercial containers normally used for that purpose.

5.2 Packing.

5.2.1 *Level A.*

5.2.1.1 The 1-gallon multifriction top cans shall be packed in accordance with the overseas equipment requirements of the appendix to Specification PPP–C–96.

5.2.1.2 The 5-gallon lug cover pails will require no packing.

5.2.2 *Level B.*

5.2.2.1 The 1-gallon multifriction top cans shall be packed in accordance with the domestic shipment requirements of the appendix to Specification PPP–C–96.

5.2.2.2 The 5-gallon lug cover pails will require no packing.

5.2.3 *Level C.* The adhesive shall be packed in a manner to insure carrier acceptance and safe delivery to destination. Containers shall comply with the Uniform Freight Classification Rules or regulations of other carriers as applicable to the mode of transportation.

5.3 Marking. In addition to any marking required by the contract or order, interior and exterior containers shall be marked in accordance with Standard MIL–STD–129 and the requirements of the Interstate Commerce Commission Regulations when applicable. Interior and exterior containers shall also be marked with the date of manufacture.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Selection of applicable levels of packaging and packing required (see 5.1 and 5.2).

6.2 With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in Qualified Products List QPL–3316, whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Chief of the Bureau of Ships, Department of the Navy, Washington 25, D.C.,

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and information pertaining to qualification of products may be obtained from that activity.

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Preparing activity:
Navy—Bureau of Ships
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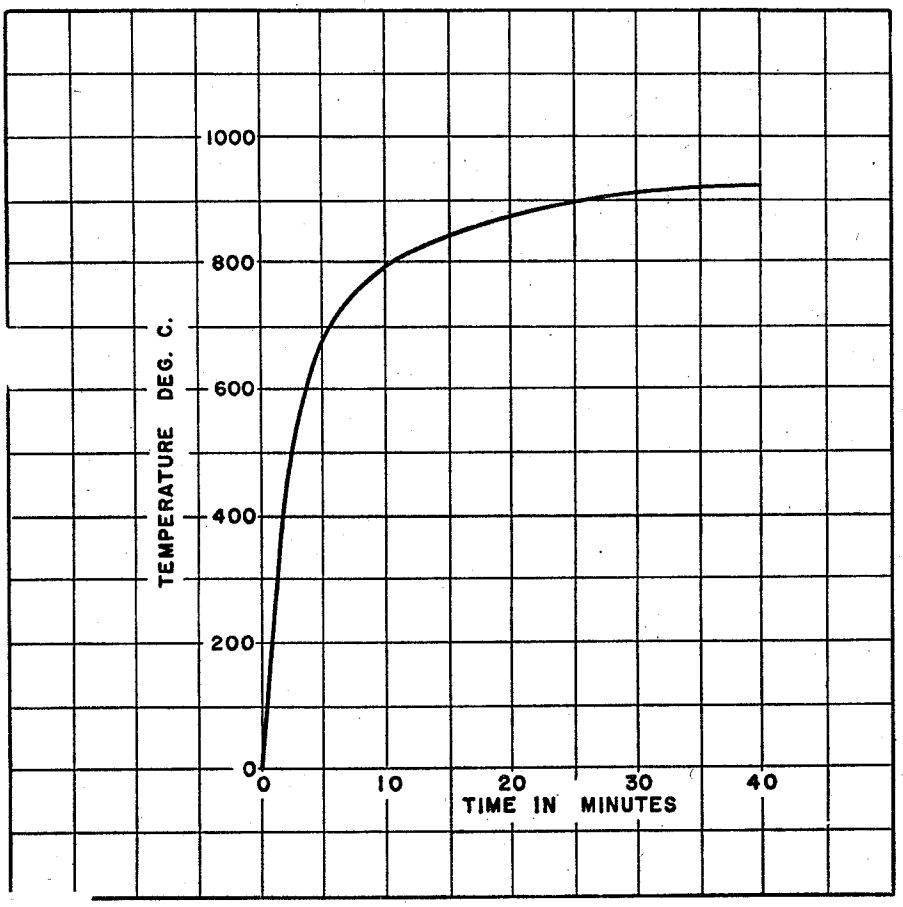


FIGURE 1. *Time temperature curve.*

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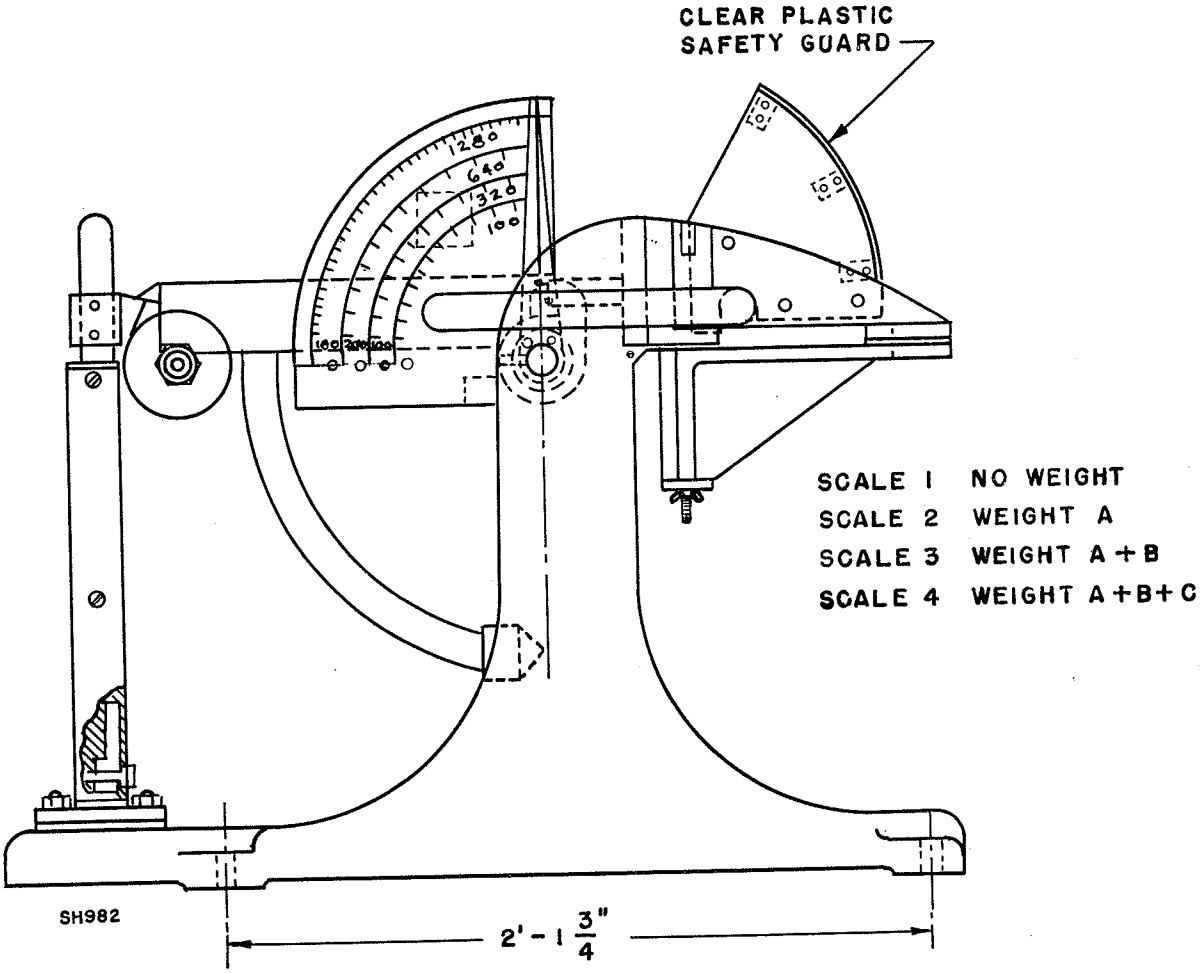


FIGURE 2. *Apparatus for puncture test.*

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